Date	28 October 2026	Team ID	EN25CS3011166
Project Name	Health360 Analytics & Risk Prediction	Maximum Marks	1 Mark

Scalability & Future Plan

Scalability & Future Plan:
This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	Static Tableau Public Visualization Data Refresh	Real-time patient monitoring and update latency	High
2	Single-node Data Storage Capacity	Restricted to static healthcare datasets; scalability limitations on larger records	Medium
3	Basic Risk Score Calculations	Lack of multi-variate predictive Machine Learning model integration	High

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Handles single-session analytics view	Deploy cloud server infrastructure with auto-scaling load balancers
2	Data Storage	Local / CSV dataset integration	Migrate to distributed cloud database (PostgreSQL / AWS Redshift)
3	Performance	In-browser Tableau render	Implement Redis caching layer and optimized query indexing
4	Security	Standard public workspace	Implement HIPAA-compliant role-based access control (RBAC) & encryption

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Real-time Patient Data Integration via REST APIs	Q1 2027	Continuous Risk Monitoring
Phase 2	Predictive Health ML Pipeline (Cardiac & Stroke Risk Scoring)	Q2 2027	Early Diagnosis Alerts
Phase 3	Cross-platform Mobile & Web Dashboard (Health360 App)	Q3 2027	Enhanced Patient Engagement
Phase 4	AI Health Assistant & Automated EHR Report Generator	Q4 2027	Clinical Decision Support

Analytics Dashboard Reference (Health360 Integration)

The upcoming phase integrates interactive analytical dimensions including:

- **Alcohol & Smoking Risk Profile:** Correlating habit frequency with cardiac stress indicators.
- **Cholesterol & Blood Pressure Variance:** Analyzing threshold impacts on overall heart disease risk.
- **BMI & Glucose Index Monitoring:** Mapping lifestyle metrics against diabetic and cardiovascular scores.